

Paper 2 - Set 09 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	A	Trigonometric functions and equations
2	E	Circles and circle theorems
3	B	Exponentials and logarithms
4	A	Identifying errors in proofs
5	C	Simultaneous equations and inequalities
6	E	Chains of reasoning
7	A	Identifying errors in proofs
8	G	Ratio, proportion and percentages
9	D	The logic of arguments
10	D	Mathematical proof
11	B	The logic of arguments
12	D	Pythagoras, mensuration and 3D
13	B	Mathematical proof
14	E	Chains of reasoning
15	B	Simultaneous equations and inequalities
16	E	Chains of reasoning
17	G	Graphs of functions
18	C	Probability
19	H	Polynomials, factor and remainder theorems
20	H	The logic of arguments

Worked solutions

Question 1 TRG-06 · Trigonometric functions and equations · spec MM4.5

Given that $\sin \theta = \frac{3}{5}$ and θ is obtuse, find $\tan \theta$.

Answer: A

$\cos \theta = -\frac{4}{5}$ (negative in the second quadrant), so $\tan \theta = \frac{3/5}{-4/5} = -\frac{3}{4}$.

Full method - how and why

From one ratio to another, with a quadrant condition fixing the signs. Use a right-angled triangle plus CAST.

1 From $\sin \theta = \frac{3}{5}$, the reference triangle has opposite 3 and hypotenuse 5, so the third side is 4 (a 3-4-5 triangle).
why: Pythagoras gives the missing side; the 3-4-5 triple appears constantly.

2 θ is obtuse, so it lies in the second quadrant where $\sin$ is positive but $\cos$ and $\tan$ are negative.

why: CAST: only sine is positive in the second quadrant.

3 So $\cos \theta = -\frac{4}{5}$ and $\tan \theta = \frac{\sin \theta}{\cos \theta} = \frac{3/5}{-4/5} = -\frac{3}{4}$.
why: The fives cancel; the sign comes from the quadrant.

Where it goes wrong

- Ignoring 'obtuse' and answering $\frac{3}{4}$.
- Inverting the ratio and answering $-\frac{4}{3}$.

Takeaway Build the reference triangle from the given ratio, then let the quadrant decide the signs.

Question 2 CIR-15 · Circles and circle theorems · spec MM3.2

A circle passes through (0, 0), (6, 0) and (0, 8). Find the coordinates of its centre.

Answer: E

The triangle has a right angle at the origin, so (6, 0)-(0, 8) is a diameter; the centre is its midpoint (3, 4).

Full method - how and why

Three points determine a circle, but here two of the chords are along the axes - and the right angle at the origin gives the centre instantly.

1 Notice the triangle (0, 0), (6, 0), (0, 8) has a right angle at the origin.

why: The two legs lie along the axes, so the angle between them is 90° .

2 A right angle inscribed in a circle stands on a diameter, so the hypotenuse from (6, 0) to (0, 8) is a diameter.

why: This is the converse of the angle-in-a-semicircle theorem.

3 The centre is the midpoint of that diameter: $(\frac{6+0}{2}, \frac{0+8}{2}) = (3, 4)$.

why: Centre is the midpoint of any diameter.

Where it goes wrong

- Averaging all three points to get $(2, \frac{8}{3})$ - that is the centroid, not the circumcentre.
- Taking the midpoint of the wrong pair, e.g. (3, 0).

Takeaway If three given points form a right angle, the hypotenuse is a diameter and its midpoint is the centre.

Question 3 LOG-15 · Exponentials and logarithms · spec MM5.2

Solve $\log_2(x^2 - 4) - \log_2(x - 2) = 3$.

Answer: B

For $x > 2$: $\log_2 \frac{(x-2)(x+2)}{x-2} = \log_2(x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6$.

Full method - how and why

Combine the logs; the algebraic fraction simplifies dramatically because the numerator factorises.

1 Combine: $\log_2 \frac{x^2 - 4}{x - 2} = 3$.

why: Difference of logs is the log of the quotient.

2 Factorise and cancel: $\frac{(x-2)(x+2)}{(x-2)(x+2)} = x+2$ (valid since $x \neq 2$).

why: The domain requires $x > 2$ for $\log_2(x-2)$ to exist, so cancelling is legitimate.

3 Convert: $\log_2(x+2) = 3 \Rightarrow x+2 = 8 \Rightarrow x = 6$.

why: $2^3 = 8$.

4 Check the domain: $x = 6$ gives $x^2 - 4 = 32 > 0$ and $x - 2 = 4 > 0$ ✓.

why: Always verify against the original.

Where it goes wrong

- Forgetting the domain and including a spurious negative solution.
- Cancelling incorrectly to get $x - 2$ or $x^2 - 2$.

Takeaway Factorise before cancelling inside a log, and confirm every solution keeps all arguments positive.

Question 4 ERR-09 · Identifying errors in proofs · spec Err2

A student proves that the equation $x^2 + 2x + 3 = 0$ has no real solutions.

I Complete the square: $x^2 + 2x + 3 = (x+1)^2 + 2$.

II $(x+1)^2 \geq 0$ for all real x .

III So $(x+1)^2 + 2 \geq 2 > 0$ for all real x .

IV Hence the equation has no real solutions.

Which of the following is correct?

Answer: A

Every step is valid.

Full method - how and why

Another valid proof. Completing the square and bounding below is a standard, correct technique.

1 Line I: $(x+1)^2 = x^2 + 2x + 1$, so $x^2 + 2x + 3 = (x+1)^2 + 2$ ✓.

why: Half the coefficient of x is 1; correct by $3 - 1 = 2$.

2 Line II: any real square is ≥ 0 ✓.

why: True for every real x .

3 Line III: adding 2 gives ≥ 2 , and $2 > 0$ ✓.

why: Adding a constant to both sides of an inequality is always valid.

4 Line IV: an expression that is always at least 2 can never equal 0, so there are no real roots ✓.

why: That is the definition of having no real solutions. (Check: discriminant $4 - 12 = -8 < 0$ ✓.)

Where it goes wrong

- Claiming $(x+1)^2$ can be negative - not for real x .
- Claiming 'a positive expression can still equal zero' - it cannot; that is what positive means.
- Assuming there must be an error.

Takeaway Completing the square gives a rigorous lower bound; if the minimum is positive, there are no real roots.

Question 5 X2-14 · Simultaneous equations and inequalities · spec MM1.5

x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?

Answer: C

$x^2 < 4$ means $-2 < x < 2$, which implies $x < 2$ (sufficient); but $x = -5$ satisfies Q and not P (not necessary).

Full method - how and why

Solve the quadratic inequality properly, then compare the two solution sets as intervals.

1 $x^2 < 4$ means $|x| < 2$, i.e. $-2 < x < 2$.

why: A quadratic inequality of this shape gives an interval **between** the roots.

2 Every x in $(-2, 2)$ satisfies $x < 2$, so $P \Rightarrow Q$: **sufficient**.

why: $(-2, 2)$ is contained in $(-\infty, 2)$.

3 Is it necessary? Take $x = -5$: $x < 2$ holds but $x^2 = 25 < 4$. So $Q \Rightarrow P$: **not necessary**.

why: The larger set contains points outside the smaller one.

4 So P is sufficient but not necessary.

why: Whenever one solution set sits strictly inside another, the inner condition is sufficient but not necessary.

Where it goes wrong

- Solving $x^2 < 4$ as $x < 2$ and concluding the conditions are equivalent - that drops the lower bound.
- Solving it as $x < -2$ or $x > 2$, which is the solution of $x^2 > 4$.

Takeaway Compare conditions as **sets**: a subset is sufficient for the superset, the superset necessary for the subset.

Question 6 RSN-21 · Chains of reasoning · spec Prf5

Seven consecutive integers are written down. Which of the following statements is/are true?

I Their sum is divisible by 7.

II Their product is divisible by 7.

III Their sum is odd.

Answer: E

The sum is $7m$ where m is the middle integer (I). Seven consecutive integers cover every remainder mod 7, so one is a multiple of 7 (II). $7m$ is odd only when m is odd, so III need not hold.

Full method - how and why

Seven consecutive integers: the sum telescopes about the middle, and the seven values cover all remainders mod 7.

1 Let the middle be m ; the seven are $m - 3, \dots, m + 3$, and the offsets cancel, so the sum is $7m$. **I is true**.

why: Symmetric representation makes the cancellation immediate.

2 II: seven consecutive integers take all seven remainders on division by 7, so exactly one is a multiple of 7; hence 7 divides the product. **True**.

why: Among any k consecutive integers there is exactly one multiple of k .

3 III: the sum is $7m$, which is odd only when m is odd. Take 1, ..., 7: sum 28, even. **Not always true**.

why: One counterexample settles it.

4 So I and II.

why: Match to the list.

Where it goes wrong

- Accepting III after testing 2, ..., 8 (sum 35, odd) and no other case.
- Confusing the sum and product claims - I is about $7m$, II about a multiple of 7 among the factors.

Takeaway k consecutive integers: their sum is $k \times$ middle (for odd k), and exactly one of them is a multiple of k .

Question 7 ERR-12 · Identifying errors in proofs · spec Err2

A student solves $x^2 = 4x$.

I Divide both sides by x : $x = 4$.

II So the only solution is $x = 4$.

Which of the following is correct?

Answer: A

Factorise instead: $x(x - 4) = 0$ gives $x = 0$ or $x = 4$.

Full method - how and why

Dividing an equation by a variable destroys the root where that variable is zero.

1 Line I divides $x^2 = 4x$ by x . That is only legitimate when $x \neq 0$ - and $x = 0$ does satisfy the equation. **First error.**

why: $0^2 = 4 \times 0$ ✓, so $x = 0$ is a genuine solution that the division removes.

2 The correct method: bring everything to one side and factorise. $x^2 - 4x = 0$, so $x(x - 4) = 0$.

why: Factorising preserves every root, because a product is zero exactly when a factor is.

3 So $x = 0$ or $x = 4$ - two solutions.

why: Line II's 'only solution' is wrong as a consequence of line I.

Where it goes wrong

- Blaming line II - it faithfully reports what line I produced; the first error is the division.
- Thinking $x^2 \div x = 2x$; it is x .
- Trying to 'fix' it by remembering to add $x = 0$ afterwards - correct in this case, but factorising is the reliable habit.

Takeaway Never divide an equation by a variable expression. Factorise instead: division loses roots, squaring adds them.

Question 8 RAT-15 · Ratio, proportion and percentages · spec M3.8

Which of the following statements is/are true?

- I Increasing a quantity by 50% and then decreasing the result by 50% returns it to its original value.
- II A 10% increase followed by a 10% increase is a 21% increase overall.
- III If A is 25% more than B , then B is 20% less than A .

Answer: G

I is false: $1.5 \times 0.5 = 0.75$. II: $1.1^2 = 1.21$, true. III: $A = 1.25B \Rightarrow B = 0.8A$, true.

Full method - how and why

Three percentage claims. Each is settled by converting to multipliers.

- 1 I: +50% then -50% gives $1.5 \times 0.5 = 0.75$, i.e. 75% of the original. **False.**

why: The decrease acts on the larger amount, so it removes more than the increase added.

- 2 II: +10% twice gives $1.1^2 = 1.21$, a 21% increase. **True.**

why: The extra 1% is the interest on the first year's interest.

- 3 III: $A = 1.25B$ means $B = \frac{A}{1.25} = 0.8A$, which is 20% less than A . **True.**

why: Percentage changes are not symmetric because the base changes.

- 4 II and III only.

why: Match to the options.

Where it goes wrong

- Accepting I because the percentages 'cancel'.
- Rejecting III on the grounds that 25% up should mean 25% down.

Takeaway Always convert to multipliers: percentage changes are asymmetric because the base changes.

Question 9 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement 'every student in the class scored more than 50'?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Full method - how and why

Negating a universal statement. 'All' flips to 'at least one', and the property inside flips too.

- 1 Write it formally: 'for all students s , $\text{mark}(s) > 50$ '.

why: Formalising stops the English from misleading you.

- 2 Negate: 'there exists a student s with $\text{mark}(s) \not> 50$ '.

why: $\neg \forall x P(x) \equiv \exists x \neg P(x)$; the quantifier flips and the inside is negated.

- 3 Translate $\not> 50$ into ' ≤ 50 ', i.e. 50 or less.

why: The negation of 'greater than' is 'less than or equal to' - the boundary belongs to the negation.

- 4 So: at least one student scored 50 or less.

why: One failure is all it takes to make 'every' false.

Where it goes wrong

- Over-negating to 'every student scored 50 or less' - that is far stronger than the negation and is not implied.
- Writing 'less than 50' and losing the case of exactly 50.

Takeaway Negating 'all ... are' gives 'at least one ... is not' - and remember $\neg(x > a)$ is $x \leq a$.

Question 10 PRF-07 · Mathematical proof · spec Prf1

Prove by cases that $n^3 - n$ is divisible by 3 for every integer n . Which set of cases is both sufficient and necessary to consider?

Answer: D

$n^3 - n = (n - 1)n(n + 1)$ is a product of three consecutive integers; splitting by remainder on division by 3 shows one factor is a multiple of 3 in every case.

Full method - how and why

Proof by exhaustive cases: the cases must cover every integer (sufficient) with none redundant (necessary), and they must be chosen to match the divisor in question.

- 1 The claim is about divisibility by 3, so split by remainder on division by 3: $n = 3k, 3k + 1, 3k + 2$.
why: Every integer has exactly one of these three forms, so the cases are exhaustive and non-overlapping.
- 2 Check each: $n^3 - n = (n - 1)n(n + 1)$. If $n = 3k$ the middle factor is a multiple of 3; if $n = 3k + 1$ then $n - 1 = 3k$; if $n = 3k + 2$ then $n + 1 = 3(k + 1)$.
why: Three consecutive integers always include a multiple of 3 - the case split makes that explicit.
- 3 So the product is divisible by 3 in every case.
why: The proof is complete because the cases cover all integers.

Where it goes wrong

- Splitting into even/odd - that is the wrong modulus and tells you nothing about divisibility by 3.
- Using only $3k + 1$ and $3k + 2$, which omits the multiples of 3 themselves - not exhaustive.
- Splitting by sign or by primality: exhaustive, but useless for this claim.

Takeaway For divisibility by m , split into the m residue classes $mk, mk + 1, \dots, mk + (m - 1)$.

Question 11 LGC-22 · The logic of arguments · spec Arg2

For real numbers a, b : P : ' $ab = 0$ '; Q : ' $a = 0$ '. Which one of the following is true?

Answer: B

$a = 0 \Rightarrow ab = 0$, so P is necessary for Q . But $ab = 0$ could be because $b = 0$ with $a \neq 0$, so P is not sufficient.

Full method - how and why

This one reverses the usual direction - the check must be done carefully rather than by pattern-matching.

- 1 Test ' P sufficient for Q ', i.e. $ab = 0 \Rightarrow a = 0$: take $a = 5, b = 0$. Then $ab = 0$ but $a \neq 0$. **Not sufficient.**
why: The zero-product rule gives $a = 0$ or $b = 0$, not $a = 0$.
- 2 Test ' P necessary for Q ', i.e. $Q \Rightarrow P$: if $a = 0$ then $ab = 0 \cdot b = 0$. **True, so necessary.**
why: Anything times zero is zero.
- 3 So P is necessary but not sufficient for Q .
why: Note the direction: the **weaker** condition ($ab = 0$) is necessary for the stronger one ($a = 0$).

Where it goes wrong

- Answering 'sufficient but not necessary' by reflex - in this question the roles are swapped from the common pattern.
- Thinking $ab = 0$ requires both to be zero.

Takeaway $ab = 0 \Rightarrow a = 0$ or $b = 0$; and always check both directions explicitly rather than assuming the familiar answer.

Question 12 X2-29 · Pythagoras, mensuration and 3D · spec M5.16

P : 'a solid cylinder and a solid cube have equal volumes'. Q : 'the cylinder and the cube have equal surface areas'. Which one of the following is true?

Answer: D

Equal volumes say nothing about surface areas (a long thin cylinder has a large surface area), and equal surface areas say nothing about volumes. Neither implies the other.

Full method - how and why

Volume and surface area are independent quantities. Neither condition constrains the other, so test both directions.

- 1 Not sufficient: take a cube of side 1 (volume 1, surface 6) and a cylinder of volume 1 that is long and thin - say radius 0.1, height $\frac{1}{0.01\pi} \approx 31.8$, whose surface area is far larger than 6.

why: Equal volumes with wildly different shapes give wildly different surface areas.

- 2 Not necessary: two solids can share a surface area while having different volumes, by the same flexibility of shape.

why: Stretching a cylinder thin at fixed surface area shrinks its volume.

- 3 So P is neither necessary nor sufficient for Q .

why: Match to the list.

Where it goes wrong

- Assuming 'bigger volume means bigger surface area' - among different shapes, no such rule holds.
- Testing only one direction and then guessing the other.

Takeaway Volume and surface area are independent across shapes; a sphere minimises surface area for a given volume, and everything else can be made worse without limit.

Question 13 PRF-13 · Mathematical proof · spec Prf1

Which of the following is a correct proof that **the sum of any three consecutive integers is divisible by 3**?

Answer: B

Writing the three as $n - 1, n, n + 1$ gives the sum $3n$ for every integer n - general and correct. Checking $1 + 2 + 3, 2 + 3 + 4, 3 + 4 + 5$ is only cases; $n, 2n, 3n$ are not consecutive; and 'any three integers include a multiple of 3' is invalid reasoning (containing a multiple of 3 does not make the sum one).

Full method - how and why

Choosing the right algebraic representation is most of the work. Centre the variables on the middle term.

- 1 Represent three consecutive integers as $n - 1, n, n + 1$.

why: Centring on n makes the ± 1 cancel; $n, n + 1, n + 2$ also works but gives $3n + 3$, needing one more line.

- 2 Add: $(n - 1) + n + (n + 1) = 3n$.

why: The -1 and $+1$ cancel.

- 3 $3n$ is 3 times an integer, hence divisible by 3, for every integer n .

why: General, so it is a proof.

Where it goes wrong

- Using $n, 2n, 3n$ - those are multiples, not consecutive integers.
- Checking three numerical cases.
- Arguing 'any three integers include a multiple of 3' - false for 1, 2, 4, and in any case containing a multiple of 3 would not make the **sum** a multiple of 3.
- The 'mean' argument that ends by demanding the middle integer be a multiple of 3 - it confuses ' $3 \times$ an integer' with ' $3 \times$ a multiple of 3'.

Takeaway Consecutive integers: use $n - 1, n, n + 1$. Their sum is $3n$, always divisible by 3.

Question 14 RSN-04 · Chains of reasoning · spec Prf5

x and y are positive integers with $x^2 - y^2 = 45$. How many possible pairs (x, y) are there?

Answer: E

$(x - y)(x + y) = 45$ with $0 < x - y < x + y$: $(1, 45), (3, 15), (5, 9)$ give $(x, y) = (23, 22), (9, 6), (7, 2)$. Three pairs.

Full method - how and why

A difference of two squares turns a quadratic condition into a factor-pair search.

1 Factorise: $x^2 - y^2 = (x - y)(x + y) = 45$.

why: Difference of two squares converts the equation into a product of two integers.

2 Both factors are positive with $x - y < x + y$, so list factor pairs of 45 with the smaller first: $(1, 45), (3, 15), (5, 9)$.

why: x, y positive forces $x + y > 0$, and then $x - y > 0$ too since the product is positive.

3 Solve each: $x = \frac{\text{sum}}{2}$, $y = \frac{\text{difference}}{2}$, giving $(23, 22), (9, 6), (7, 2)$.

why: Adding the two equations gives $2x$; subtracting gives $2y$. Both factors are odd, so both results are integers.

4 All three have positive integer y , so there are **three** pairs.

why: Check one: $9^2 - 6^2 = 81 - 36 = 45 \checkmark$.

Where it goes wrong

- Including $(9, 5)$ or other reversed pairs - $x - y$ must be the smaller factor.
- Forgetting to check that x and y come out as integers; if 45 had a factor pair of mixed parity it would have to be discarded.
- Counting $45 = 45 \times 1$ and 1×45 as two different solutions.

Takeaway $x^2 - y^2 = N$: list factor pairs $N = de$ with $d < e$ and the same parity, then $x = \frac{d+e}{2}$, $y = \frac{e-d}{2}$.

Question 15 SIM-18 · Simultaneous equations and inequalities · spec MM1.5

Consider the statement: for all real x , if $x > 2$ then $x^2 - 3x + 2 > 0$. Which of the following is true?

Answer: B

$x^2 - 3x + 2 = (x - 1)(x - 2) > 0$ exactly when $x < 1$ or $x > 2$. So $x > 2$ does imply it (true), but the converse fails, e.g. $x = 0$ gives $2 > 0$ without $x > 2$.

Full method - how and why

A Paper 2-style wrapper on a quadratic inequality: decide the statement and its converse separately.

1 Factorise the quadratic: $x^2 - 3x + 2 = (x - 1)(x - 2)$, which is positive exactly when $x < 1$ or $x > 2$.

why: Knowing precisely where the expression is positive lets you test both directions quickly.

2 The statement: does $x > 2$ guarantee positivity? Yes - $x > 2$ lies inside the region $x > 2$. **True.**

why: An implication is true when the hypothesis region sits inside the conclusion region.

3 The converse: does positivity guarantee $x > 2$? No - $x = 0$ gives $(0 - 1)(0 - 2) = 2 > 0$ but $0 \not> 2$. **False.**

why: The positivity region is strictly larger than $x > 2$, so a counterexample must come from the extra part $x < 1$.

4 So the statement is true but its converse is false.

why: Report both verdicts in the order the options use them.

Where it goes wrong

- Assuming the converse holds because the implication does - converses are logically independent.
- Only testing values greater than 2, which can never refute the converse.

Takeaway An implication is true when one region is contained in the other; its converse needs the reverse containment, so look for a counterexample in the leftover region.

Question 16 X2-26 · Chains of reasoning · spec Prf5

The positive integers a , b and c satisfy $a < b < c$ and $a + b + c = 12$. Which of the following statements must be true?

- I $a \leq 3$
- II $c \geq 5$
- III $b = 4$

Answer: E

Since $a < b < c$, $3a < 12$ so $a \leq 3$ (I); $3c > 12$ so $c \geq 5$ (II). III fails: $1 + 2 + 9 = 12$ has $b = 2$.

Full method - how and why

Extremal reasoning: squeeze a and c using the ordering, then test the remaining claim with an example.

1 I: since $a < b < c$, we have $3a < a + b + c = 12$, so $a < 4$, i.e. $a \leq 3$ for an integer. **True.**

why: Replacing b and c by the smallest value a gives a strict under-estimate of the sum.

2 II: similarly $3c > a + b + c = 12$, so $c > 4$, i.e. $c \geq 5$. **True.**

why: Replacing a and b by the largest value c over-estimates the sum.

3 III: must $b = 4$? Try $1 + 2 + 9 = 12$ with $1 < 2 < 9$ ✓ and $b = 2$. **False.**

why: One valid triple with $b \neq 4$ settles it.

4 So I and II.

why: Match to the list.

Where it goes wrong

- Writing $3a \leq 12$ instead of $3a < 12$; the strict ordering $a < b < c$ gives a strict inequality, though here both routes end at $a \leq 3$.
- Assuming b must be the mean of the three - true only for equally spaced values.

Takeaway Bound an ordered sum by replacing every term with the smallest (or largest): $3a < a + b + c < 3c$.

Question 17 X2-21 · Graphs of functions · spec MM8.6

$p(x)$ is a polynomial of degree 4 with real coefficients. Which of the following statements must be true?

- I $p(x) = 0$ has at least one real root.
- II $p(x) = 0$ has at most four real roots.
- III The graph of $y = p(x)$ has at least one turning point.

Answer: G

I fails: $x^4 + 1$ has no real roots. II: a degree-4 polynomial has at most 4 roots. III: an even-degree polynomial tends to the same infinity at both ends, so it must turn at least once.

Full method - how and why

Three claims about a quartic. Two follow from degree; the first needs a counterexample.

1 I: must a degree-4 polynomial have a real root? No: $x^4 + 1 \geq 1 > 0$ for all real x . **False.**
why: Even-degree polynomials can stay entirely above the axis - unlike odd-degree ones.

2 II: a polynomial of degree n has at most n real roots, so at most 4. **True.**
why: Each root supplies a linear factor; more than 4 would exceed the degree.

3 III: as $x \rightarrow \pm \infty$, a quartic tends to the same infinity at both ends (the sign of the leading coefficient).
why: Even powers dominate and are positive at both extremes.

4 A continuous function going to the same infinity both ways must turn round somewhere, so it has at least one turning point. **True.**
why: It has a global minimum (or maximum) at a finite point.

5 So II and III.
why: Match to the list.

Where it goes wrong

- Accepting I by analogy with odd-degree polynomials, which **must** have a real root.
- Reading II as 'exactly four roots' - repeated roots and non-real roots reduce the count.

Takeaway Odd degree guarantees a real root; even degree guarantees a turning point. Degree n caps the real roots at n .

Question 18 X2-19 · Probability · spec M7.7

Events A and B have $P(A) > 0$ and $P(B) > 0$. P : ' A and B are mutually exclusive'. Q : ' A and B are not independent'. Which one of the following is true?

Answer: C

Mutually exclusive gives $P(A \text{ and } B) = 0 \neq P(A)P(B)$, so they cannot be independent: P is sufficient. But two overlapping events can also fail to be independent, so P is not necessary.

Full method - how and why

Relating mutual exclusivity to independence. One implication holds; the converse has easy counterexamples.

- 1 If A, B are mutually exclusive, $P(A \cap B) = 0$. But $P(A)P(B) > 0$ since both probabilities are positive.
why: The positivity hypothesis is what makes the argument work.
- 2 So $P(A \cap B) \neq P(A)P(B)$: they are **not** independent. $P \Rightarrow Q$: **sufficient**.
why: Independence is exactly the equality that fails here.
- 3 Necessity: find dependent events that are **not** mutually exclusive. Roll a die with A : 'even' and B : 'at most 3'.
Then $P(A \cap B) = P(\{2\}) = \frac{1}{6}$, but $P(A)P(B) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$.
why: Unequal, so they are dependent - yet they overlap, so they are not mutually exclusive.
- 4 So P is sufficient but not necessary.
why: Match to the list.

Where it goes wrong

- Treating mutually exclusive and independent as opposites in a way that makes them necessary for each other - dependence has many other causes.
- Forgetting the hypothesis $P(A), P(B) > 0$; with a zero probability, events can be both mutually exclusive and independent.

Takeaway For events of positive probability, mutually exclusive $\Rightarrow$ dependent, but not conversely.

Question 19 POL-08 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following statements about the polynomial $p(x) = x^3 - 6x^2 + 11x - 6$ is/are true?

- I** $p(x)$ has three distinct real roots.
- II** $(x - 4)$ is a factor of $p(x) - 6$.
- III** $p(x) > 0$ for all $x > 3$.

Answer: H

$p(x) = (x - 1)(x - 2)(x - 3)$, so I is true. $p(4) - 6 = 3 \cdot 2 \cdot 1 - 6 = 0$, so II is true. For $x > 3$ all three factors are positive, so III is true.

Full method - how and why

Three statements about one cubic. Factorising it first makes all three immediate.

- 1** Factorise: $p(1) = 1 - 6 + 11 - 6 = 0$, so $(x - 1)$ is a factor and $p(x) = (x - 1)(x - 2)(x - 3)$.
why: Test the divisors of 6; the resulting quadratic $x^2 - 5x + 6$ factorises easily.
- 2** I: the roots 1, 2, 3 are distinct. **True.**
why: Three distinct linear factors means three distinct roots.
- 3** II: $(x - 4)$ is a factor of $p(x) - 6$ iff $p(4) - 6 = 0$. Now $p(4) = 3 \times 2 \times 1 = 6$, so $p(4) - 6 = 0$. **True.**
why: Factor Theorem applied to the shifted polynomial - evaluate the whole expression $p(x) - 6$ at $x = 4$.
- 4** III: for $x > 3$ each factor $x - 1$, $x - 2$, $x - 3$ is positive, so the product is positive. **True.**
why: Sign analysis from the factorised form; beyond the largest root everything is positive.
- 5** All three: 'I, II and III'.
why: Match verdicts to options.

Where it goes wrong

- Expanding $p(x) - 6$ and re-testing from scratch instead of just evaluating $p(4) - 6$.
- Checking III at one value like $x = 10$ and calling it proved - the factor argument covers every $x > 3$.

Takeaway Factorise first: roots, sign and shifted-factor questions then all read off the same form.

Question 20 LGC-09 · The logic of arguments · spec Arg1

Consider the statement S : **for all real x , if $x > 0$ then $x + \frac{1}{x} \geq 2$** . Which of the following statements is/are true?

- I S is true.
- II The converse of S (for all real x , if $x + \frac{1}{x} \geq 2$ then $x > 0$) is true.
- III The contrapositive of S is true.

Answer: H

For $x > 0$, $x + \frac{1}{x} - 2 = \frac{(x-1)^2}{x} \geq 0$, so S is true, and so is its contrapositive (III). For the converse: if $x < 0$ then $x + \frac{1}{x} < 0 < 2$, so $x + \frac{1}{x} \geq 2$ forces $x > 0$ (II true).

Full method - how and why

One inequality, examined three ways. The contrapositive is free once the statement is settled; the converse must be tested separately.

- 1 I: for $x > 0$, write $x + \frac{1}{x} - 2 = \frac{x^2 - 2x + 1}{x} = \frac{(x-1)^2}{x}$.

why: Put over a common denominator x - the numerator turns out to be a perfect square.

- 2 Since $(x-1)^2 \geq 0$ and $x > 0$, the fraction is ≥ 0 , so $x + \frac{1}{x} \geq 2$. **True.**

why: A non-negative numerator over a positive denominator is non-negative. Equality at $x = 1$.

- 3 III: the contrapositive of a true statement is automatically **true**.

why: No work needed - contrapositive and original are logically equivalent.

- 4 II (the converse): suppose $x + \frac{1}{x} \geq 2$; must $x > 0$? If $x < 0$ then both x and $\frac{1}{x}$ are negative, so $x + \frac{1}{x} < 0 < 2$ - the hypothesis fails. And $x = 0$ is not allowed. So $x > 0$. **True.**

why: Proved by eliminating the alternatives, which is itself a contrapositive argument.

- 5 So all three.

why: Match to the list.

Where it goes wrong

- Assuming the converse is false because converses 'usually' are - here it genuinely holds.
- Multiplying the inequality $x + \frac{1}{x} \geq 2$ by x without knowing the sign of x ; that is precisely what is in question.

Takeaway $x + \frac{1}{x} \geq 2$ for $x > 0$ is worth knowing; and a statement always shares its truth value with its contrapositive, never necessarily with its converse.